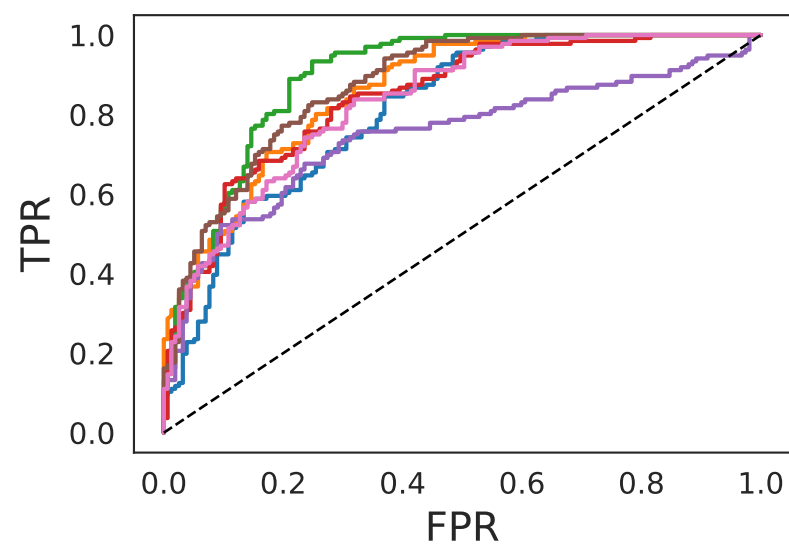
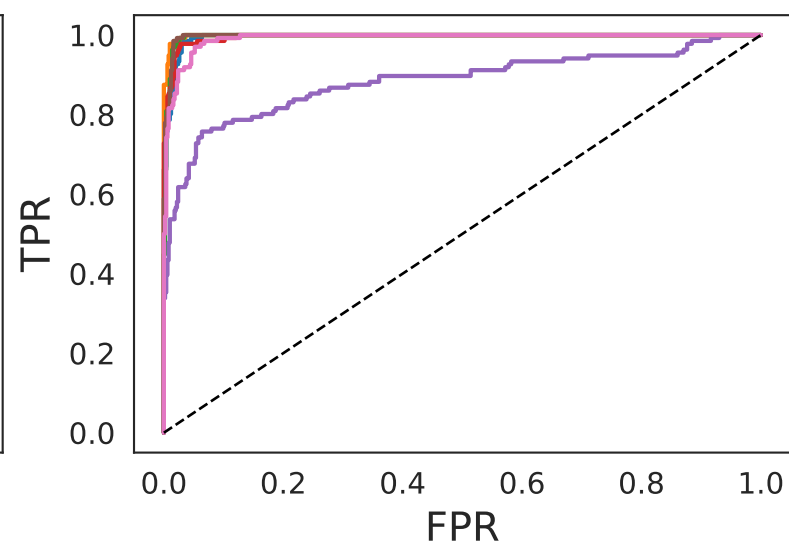


Validated tested
binders vs non-binders



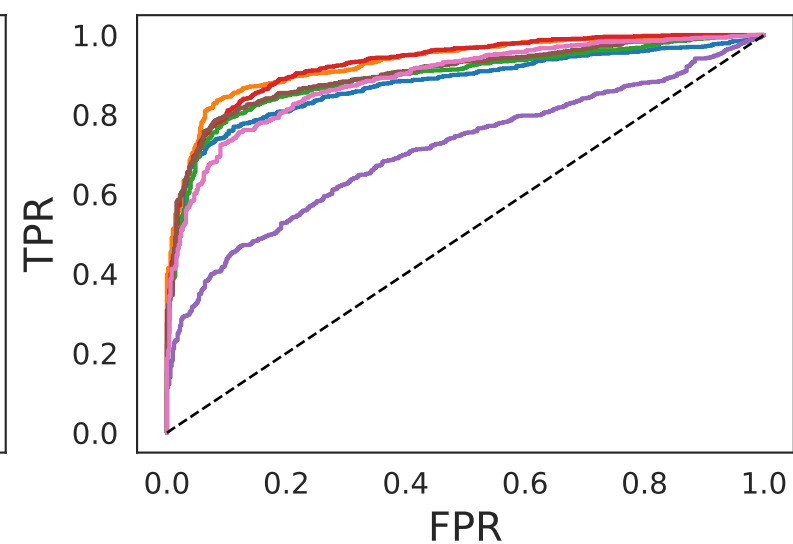
- MHCflurry (AUC=0.81)
- NetMHCpan %Rank (AUC=0.86)
- ESMCBA log IC50 (AUC=0.90)
- HLAthena %Rank (AUC=0.84)
- Apollo inverse (AUC=0.75)
- MHC Nugget (AUC=0.87)
- MixMHCpred %Rank (AUC=0.83)

Validated strong
binders vs ESM2 random



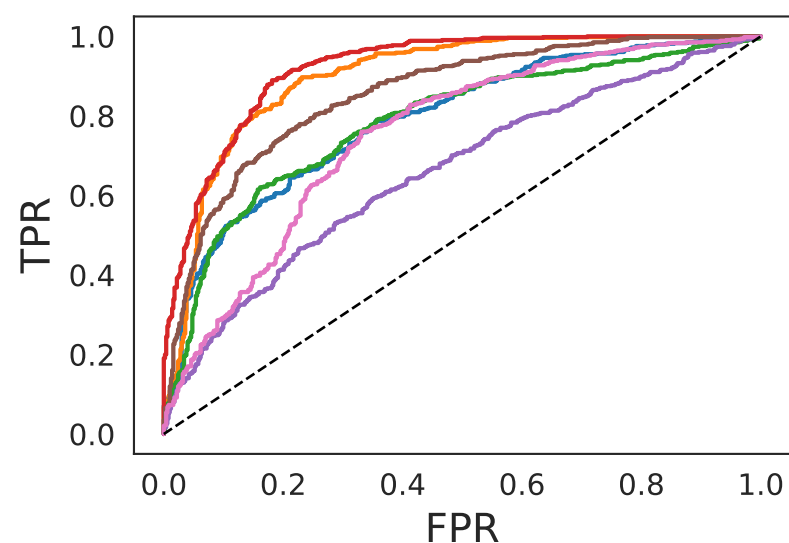
- MHCflurry (AUC=0.99)
- NetMHCpan %Rank (AUC=1.00)
- ESMCBA log IC50 (AUC=1.00)
- HLAthena %Rank (AUC=1.00)
- Apollo inverse (AUC=0.88)
- MHC Nugget (AUC=1.00)
- MixMHCpred %Rank (AUC=0.99)

Anchor-preserved
vs ESM2 random



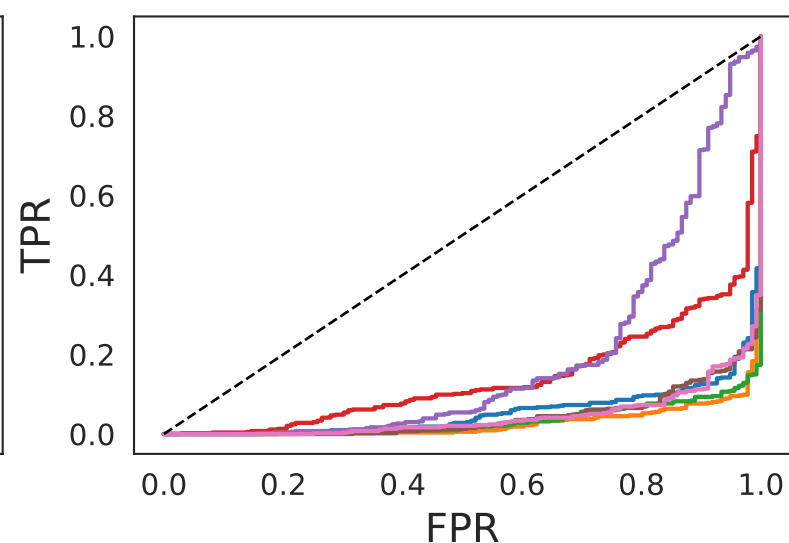
- MHCflurry (AUC=0.88)
- NetMHCpan %Rank (AUC=0.93)
- ESMCBA log IC50 (AUC=0.89)
- HLAthena %Rank (AUC=0.93)
- Apollo inverse (AUC=0.71)
- MHC Nugget (AUC=0.90)
- MixMHCpred %Rank (AUC=0.89)

Diffusion designs
vs ESM2 random



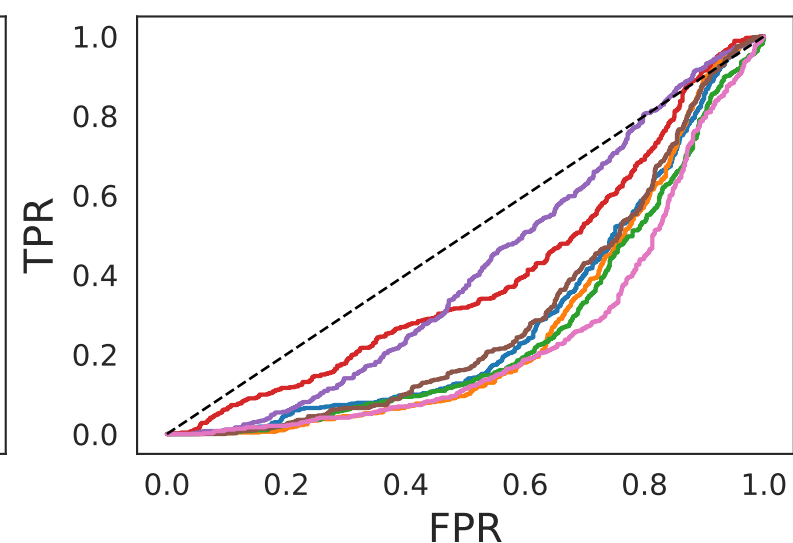
- MHCflurry (AUC=0.79)
- NetMHCpan %Rank (AUC=0.90)
- ESMCBA log IC50 (AUC=0.78)
- HLAthena %Rank (AUC=0.92)
- Apollo inverse (AUC=0.66)
- MHC Nugget (AUC=0.85)
- MixMHCpred %Rank (AUC=0.75)

Diffusion designs
vs Binders



- MHCflurry (AUC=0.05)
- NetMHCpan %Rank (AUC=0.03)
- ESMCBA log IC50 (AUC=0.03)
- HLAthena %Rank (AUC=0.14)
- Apollo inverse (AUC=0.19)
- MHC Nugget (AUC=0.04)
- MixMHCpred %Rank (AUC=0.04)

Diffusion designs
vs Anchor-preserved



- MHCflurry (AUC=0.29)
- NetMHCpan %Rank (AUC=0.27)
- ESMCBA log IC50 (AUC=0.27)
- HLAthena %Rank (AUC=0.40)
- Apollo inverse (AUC=0.42)
- MHC Nugget (AUC=0.30)
- MixMHCpred %Rank (AUC=0.24)